

# DURWAS KHOT

MSc Automotive Engineering | CAE Engineer | FEA Simulation | ANSA | OEM Experience | Python  
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## SUMMARY

Automotive Engineer with hands-on experience developing complex systems using CAE and FEA for industrial and competitive projects. Awarded for outstanding performance at Tritorc and recognised for technical leadership at Formula Student with the Best Debut award. Proven success in cross-functional environments, driving strategic outcomes for internal as well as external stakeholders.

## EDUCATION

**MSc Automotive Eng.**, Oxford Brookes University (1:1) Sept 2024 – Sept 2025  
*Top achievement: Selected for a competitive industry-sponsored thesis with Bentley Motors, delivering MBD-based NVH analysis at an industry-comparable standard and graduating with Distinction.*

**BEng Mechanical Eng.**, University of Mumbai (77.66% - UK Equivalent – 1:1) Aug 2019 - May 2023  
*Top achievement: Co-founded an FS team as Team Manager, engineering a system winning the Best Debut Award.*

## TECHNICAL SKILLS

**Functional Competencies:** Finite Element Analysis (FEA) | Structural Analysis | CAE Methodologies | Pre-processing & Model Preparation | Geometry Cleanup, Meshing & Mesh Quality Checks | Crashworthiness Simulation | Model Validation & Correlation | Strength of Materials (SOM) | Engineering Mechanics | Composite Materials | DFMEA / PFMEA | Analytical Problem-Solving | Technical Reporting | Cross-functional Collaboration

**Technical Tools:** LS-DYNA | ANSYS | ANSA | CATIA | SolidWorks (CSWA) | MSC Adams | MATLAB/Simulink | Python Programming and Automation (NumPy, Pandas, Matplotlib)

## RESEARCH AND INDUSTRY PROJECTS

**Master's Thesis Student**, Bentley Motors with Oxford Brookes May 2025 – Sep 2025  
*Relevance: CAD-integrated analysis and design validation under real OEM product development constraints*

- Developed flexible MBD models and PINN-based characterisation in Python, automating engineering calculations and data-visualisation for steering system bushes in limited availability of test data, achieving 25% improvement in bushing stiffness/damping correlation.
- Performed NVH data analysis across different test cases, delivering design insights that reduced steering VTV from 2.38 to 1.64 m/s<sup>2</sup> through non-linear bushing integration, validated against ISO 5349 comfort metrics.
- Conducted correlation of high-frequency (above 80 Hz) road-induced vibration in electric vehicle systems through test and flexible multi-body dynamics simulation to improve model prediction accuracy by 30%.

**Chassis - Composites and Impact Analysis**, Oxford Brookes University Jan 2025 – Mar 2025  
*Relevance: Structural safety design and impact-energy management for a composite monocoque vehicle chassis.*

- Led a team project for structural and stress analysis of a composite monocoque chassis for hydrogen vehicle crashworthiness using LS-DYNA FEA and hand-calculation methods.
- Designed and improved the design of front and side impact attenuators using carbon composite laminate and Nomex material, resulting in reduced deflection by 85% in frontal impact and 89% in side impact.
- Validated structural performance against 1300 kg vehicle impacts at 13.88 m/s as per Euro NCAP standards. Controlled kinetic energy absorption prevented potential hydrogen tank displacement and reinforcement failure.

## WORK EXPERIENCE

**Junior Design Engineer**, TRITORC Equipments Jun 2023 – Aug 2024  
*Relevance: End-to-end mechanical design and NPD experience on production hydraulic systems*

- Designed and validated 5+ hydraulic systems using CATIA modelling, GD&T-compliant 2D drawings and FEA for CNC/VMC manufacturing, reducing component failure rate/warranty returns.
- Applied DFMEA/PFMEA and engineering stress calculations to redesign critical hydraulic components against fatigue and operational loading, improving structural reliability by 20% while keeping material cost within target.
- Led cross-functional coordination with production and testing, including customer visits to resolve manufacturability issues, implementing lean manufacturing techniques and aligning designs with regulatory standards, reducing design review cycle time and supporting successful product launches into global markets.